LESSON 05 · CHEAT SHEET

Strings, Files & CSV

SPLIT & JOIN

```
"Robinson, Max".split(", ")   ['Robinson', 'Max']
"a b  c".split()              ['a', 'b', 'c']
"x;y;z".split(";")            ['x', 'y', 'z']
", ".join(["a", "b"])         'a, b'
text.splitlines()             # list of lines
last, first = s.split(", ")   # unpack
sorted(set(items))            # unique, sorted
```

WRITE A TEXT FILE

```
with open("my_work/out.txt", "w",
          encoding="utf-8") as f:
    f.write("one line\n")      # add \n yourself
    f.write(f"total: {x}\n")
```

`"w"`	write (replaces the file!)
`"a"`	append to the end
`"r"`	read (the default)

READ A TEXT FILE

```
with open("my_work/out.txt",
          encoding="utf-8") as f:
    text = f.read()           # whole file, one str

with open(path, encoding="utf-8") as f:
    for line in f:            # line by line
        line = line.strip()   # drop the \n

lines = f.readlines()         # list of lines
```

The `with` block closes the file for you. Indent everything that needs the file.

READ A CSV

```
import csv
with open("data/x.csv", newline="",
          encoding="utf-8") as f:
    reader = csv.DictReader(f)
    print(reader.fieldnames)  # header
    for row in reader:        # row = dict
        print(row["Salary"])  # always TEXT
    rows = list(reader)       # or keep them all
```

Values arrive as strings: `int()` / `float()` them. Like VBA's `Open … For Input` + `Split`, minus the pain.

WRITE A CSV

```
with open("my_work/clean.csv", "w",
          newline="", encoding="utf-8") as f:
    w = csv.DictWriter(f, fieldnames=["id", "name"])
    w.writeheader()
    w.writerow({"id": "EMP1", "name": "Jen"})
    w.writerows(list_of_dicts)
```

CLEANING RECIPES

```
s.strip().upper()           # " emp1 " -> "EMP1"
s.replace("£", "").replace(",", "")
float(s) if s != "" else 0.0
s.replace("hrs", "").strip()
first.strip().title()       # "AMARA" -> "Amara"
d[k] = d.get(k, 0) + v      # count into a dict
```

PATHLIB

```
from pathlib import Path
p = Path("data") / "payroll.csv"   # joins with /
p.name        'payroll.csv'
p.exists()    True
p.read_text(encoding="utf-8")      # whole file
p.write_text("hi\n", encoding="utf-8")
for p in sorted(Path("data").glob("*.csv")):
    ...                            # every CSV
```

GOTCHAS

- `"w"` wipes the file before writing. Never open `data/` files with `"w"`
- CSV: always `newline=""` and `encoding="utf-8"`
- Don't `.split(",")` a CSV line: "Robinson, Max" has a comma inside the quotes
- Run scripts from the lesson folder, or `data/…` isn't found
- Forgot the `\n`? Everything lands on one line